

Invoice Processing Pipeline — Design Reference v8 (Final)

Power Automate orchestration; binary transport; Function decision engine; no duplicate-invoice verification. Supersedes v7.
Prepared: 18 June 2026 · Status: Approved for prototype · Supersedes v7

What changed since v7

v8 carries a single change from v7: the orchestrator moves from Logic Apps to Power Automate. A Power Automate cloud flow now owns the entire pipeline — the trigger, the Function call, the routing branch, the Dataverse write, and the terminal ledger update — as well as the review flows it already ran. Logic Apps is no longer used. The decision engine (the Azure Function), the gates, the binary transport, the item-id ledger, and the prebuilt-document analyzers are unchanged.

Area	v7	v8
Orchestrator	Logic Apps Standard (the spine)	Power Automate cloud flow
Power Automate scope	Review subsystem only	Entire pipeline — orchestration spine + review flows
Logic Apps	Owns the spine and the Dataverse write	Removed from the architecture
Function call (orchestrator to Function)	Logic Apps HTTP action with managed identity	Power Automate custom connector (function-key connection for the prototype; Entra federated-credential managed identity for production)
Dataverse / SharePoint (orchestrator to them)	Managed identity	Connection references (a user or service-principal connection)
Function's own auth (to CU, Table)	User-assigned managed identity	Unchanged

Why this changes the auth model. Power Automate's HTTP action does not support managed identity (that is a Logic Apps capability), so the orchestrator can no longer authenticate to the Function or to Dataverse with a managed identity the way the Logic-Apps spine did. The Function call goes through a Power Automate custom connector — the function key held in the connection for the prototype, or a secretless Entra federated-credential managed identity for production. Dataverse and SharePoint are reached through connection references. The Function's own outbound identity (to Content Understanding and Table Storage) is its user-assigned managed identity and is unaffected.

Accepted consequence. Because the orchestration spine is now Power Automate too, a Power Automate service degradation affects the whole pipeline, not just the review subsystem — the isolation the Logic-Apps spine provided is gone. Power Automate throttling, concurrency limits, and per-action request timeouts now apply to the spine. See the risks section.

Resolved design decisions

#	Question	Answer	Design impact
1	Property tax ever handwritten?	Never	is_handwritten lives only on the general invoice analyzer
2	Dynamics product?	Dynamics 365 Customer Engagement	Dataverse connector 'Add a new row'; invoice table is the target
3	Latency requirement?	5-minute interval acceptable	SharePoint polling trigger; no Blob landing zone needed
4	Verify duplicate invoices?	No (requirement)	No Dataverse query and no alternate key; A1 idempotency only
5	Multiple invoices per PDF?	No — one per PDF	No segmentation; router processes one content object per file
6	Document identity?	SharePoint item UniqueId	No byte hash; A1 is an atomic idempotency claim
7	How does CU receive the file?	Binary upload from the Function	No Blob staging area and no SAS token lifecycle
8	Where does routing logic run?	Azure Function (decision engine)	Gates and the CU call sit in Python, not in orchestrator expressions
9	Classifier confidence gate (B1)?	Not available at GA	B1 dropped; the field-confidence gate B4 is the fallback
10	Orchestrator?	Power Automate cloud flow	Trigger, Function call (custom connector), branch, Dataverse write; Logic Apps not used

1. Architecture overview

Three stages — trigger, classify-and-decide, route-and-write — with a Power Automate cloud flow as the sole orchestrator and single owner of the Dataverse write. The flow's SharePoint trigger fires within 1–5 minutes of upload. The flow fetches the file content and the item UniqueId, then calls the Azure Function decision engine through a custom connector: the Function atomically claims the item id in the ledger (gate A1, idempotency), submits the document to Content Understanding as binary, applies the routing gates, records the decision, and returns it. The flow then branches: the happy path writes the invoice to Dynamics directly; any failed gate writes a review-queue item instead. The pipeline performs no duplicate-invoice check.

Component responsibilities

Component	Role	Technology
SharePoint library	Invoice landing zone; one invoice per PDF	Microsoft 365 SharePoint
Power Automate (orchestration flow)	Trigger; file fetch; calls the Function (custom connector); Dataverse write; review-item write; terminal ledger update	Power Automate (cloud flow)
Azure Function	Decision engine: A1 atomic claim, CU call, gates B2–B4 and GST, ledger writes	Azure Functions (Python)
Content Understanding	Single call: classify, route, extract	Azure Content Understanding GA (2025-11-01)
General invoice analyzer	Fields plus is_handwritten, vendor_category, anomaly_flag	CU — prebuilt-document base, OCR on
Property tax analyzer	Custom tax schema, digital only (deferred in prototype)	CU — prebuilt-document base
InvoiceExtractProcessLog	Idempotency gate A1 (atomic claim) plus file-to-Dynamics audit ledger	Azure Table Storage
Human review queue	Low-confidence, handwritten, rejected items	SharePoint list (InvoiceReviewQueue)
Review flows + app	Notify and approval flows; reviewer app	Power Automate and Power Apps
Dynamics 365 CE	Final destination, invoice table	Dataverse connector

2. End-to-end sequence flow

From upload to the Dataverse write, on the binary transport. The Function owns everything up to the decision; the Power Automate flow owns the trigger, the branch, and all writes. Gate A1 atomically claims the SharePoint item id and short-circuits one already processed.

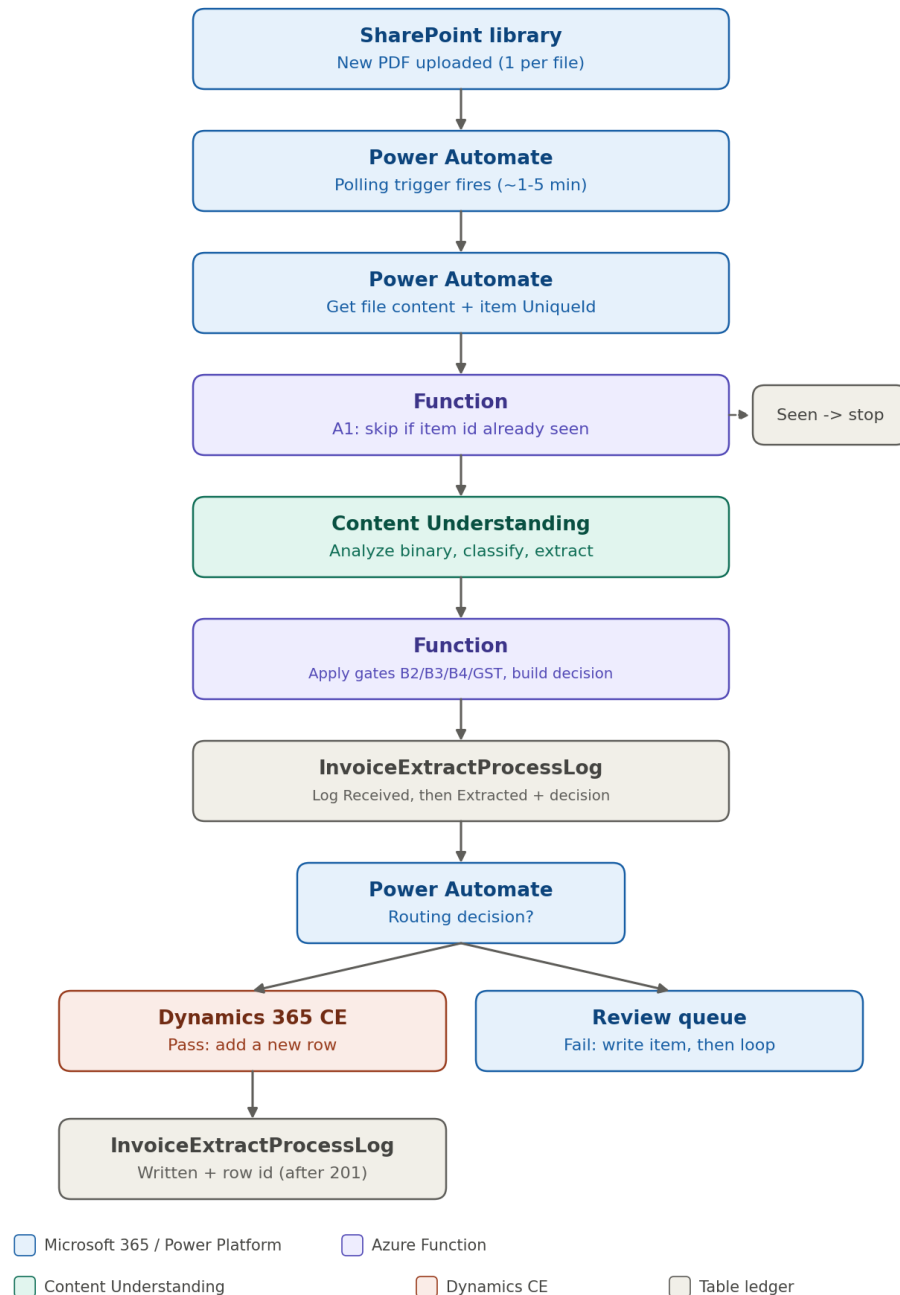


Figure 1. End-to-end sequence. Power Automate is the orchestrator; it calls the Function through a custom connector and writes to Dynamics through a Dataverse connection. Gate A1 is an atomic idempotency claim, not a duplicate-invoice check; the happy path writes directly. The ledger is written by the Function (Received, then Extracted) and finalized by the Power Automate flow (Written plus the Dynamics row id) after a 201. The review off-ramp (right) is detailed in section 7.

3. Classification taxonomy

Three business categories. Handwriting is a modality flag inside the general invoice analyzer only — property tax bills are never handwritten in this population, so the property tax schema carries no modality field. In the current prototype the router implements two categories (general_invoice and other); property tax is retained in the design and added when its custom schema is built.

Category	Description	Routed to
general_invoice	Commercial billing from vendors, trades, insurers or utilities requesting payment	General invoice analyzer
property_tax	Government property tax bills with parcel IDs, assessed values, levy breakdowns (deferred)	Property tax analyzer
other	Catch-all for any non-invoice document. Mandatory.	Reject — no extraction

Analyzer base note: the general invoice analyzer is built on prebuilt-document, not prebuilt-invoice. prebuilt-invoice is not supported as a base analyzer, and the prebuilt-document base is what allows the custom is_handwritten classify field to be added. OCR stays enabled on both analyzers.

4. Dynamics 365 Customer Engagement integration

Dynamics 365 CE is Dataverse-based. The Power Automate flow uses the Dataverse connector — not the deprecated legacy Dynamics 365 connector — and authenticates through a connection reference (a user or service-principal connection), not a managed identity. Now that duplicate invoices are not verified, only one operation is needed: 'Add a new row' to create the invoice on the happy path or after an approved review.

Operation	Dataverse action	When	Purpose
Invoice creation	Add a new row	Happy path or approved review	Create the invoice record; returns the row ID

Canonical field mapping (logical names to be confirmed by the CE administrator):

Source (general invoice)	Source (property tax)	CE invoice field (placeholder)
invoice_number	parcel_id	invoicenumber
total_invoice_amount	total_tax_due	totalamount
payment_due_date	instalment_1_due	duedate
vendor_name	tax_authority	name / vendor (account) lookup

Confirm before Phase 6. CE invoice records typically reference a vendor through an account or contact lookup, not a free-text name, so a 'find or create account' step may be needed before the invoice write — the single largest unknown in the write path. Also decide whether the flow's Dataverse connection runs as a named service account or a service principal. The alternate-key question is withdrawn (there is no duplicate-enforcement key).

5. Routing decision rules — two stages

Stage A runs inside the Function before the Content Understanding call. Stage B evaluates the CU result. Short-circuit on the first match.

Stage A — pre-extraction

Priority	Condition	Rule	Action
A1	SharePoint item id already claimed and decided in the ledger	Same upload already processed (idempotency, not duplicate-invoice)	Skip — return the stored decision; do not reprocess

Stage B — post-extraction (evaluating the CU response)

Priority	Condition	Action
B2	Category = other	Reject — no further processing
B3	is_handwritten = yes (general invoice only)	Review — modality policy, no exceptions at launch
B4	Critical field confidence below 0.75	Review — low confidence
GST	total minus GST does not reconcile to 5% (within tolerance)	Review — GST arithmetic inconsistent
B6	line_items row confidence below 0.65 (advisory)	Flag row; continue to write
B7	All conditions clear	Power Automate adds a new row to Dataverse

B1 and B5 removed. B1 (classifier confidence) cannot be implemented — the GA contentCategories response carries no per-category score, so B4 is the fallback. B5 (the Dataverse duplicate query) is removed by requirement, so a happy-path candidate goes straight to the write. B6 is advisory only and is inert for the current general invoice schema, which has no line-items array. anomaly_flag remains advisory.

6. InvoiceExtractProcessLog (Table Storage)

Keyed on the SharePoint item id. With duplicate-invoice verification gone, gate A1 is the only 'seen before' mechanism, and it is idempotency, not dedup: an atomic insert-claim ensures one upload is processed once even under concurrent trigger replays. The table also remains the processing ledger and the source of stuck-invoice visibility. It is written by the Function (Received, then Extracted) and finalized by the Power Automate flow (Written plus the Dynamics row id) after a confirmed 201.

Property	Type	Notes
PartitionKey	String	First characters of the sanitized SharePoint item id (load spread)
RowKey	String	Sanitized SharePoint item UniqueId — the A1 idempotency key
Status	String	Received / Extracted / InReview / Written / Rejected
RoutingDecision	String	Gate outcome returned by the Function
DocumentType	String	general_invoice / property_tax
IsHandwritten	String	yes / no / n a — audit trail
DynamicsRecordId	String	CE invoice row ID, populated only after a confirmed write (the design's DataverseRowId)
IngestedUtc / LastUpdatedUtc	DateTime	First sighting and most recent update
SourceFileUrl	String	SharePoint URL of the original PDF
ExpiresAt	DateTime	Drives the nightly purge; minimum 13 months retention

Stuck-invoice signal. An empty DynamicsRecordId means the write never completed — these rows are the in-flight or stuck set and are never purged. Retention follows the finance team's audit policy; Table Storage has no native TTL, so a nightly Azure Function timer removes expired rows.

7. Human review queue and Power Automate

Power Automate now runs the whole pipeline, so the review subsystem is no longer a separate platform from the spine — it is additional flows on the same one. When a gate fails, the orchestration flow writes an item to the SharePoint review list (the Function does not write to SharePoint). Two further Power Automate flows trigger off that list: a notify flow and an approval flow.

- **Notify flow** — fires on item creation; posts a Teams adaptive card with key fields and the firing gate inline.
- **Approval flow** — fires on status = Approved; re-enters the orchestration flow's write action with the reviewer's corrected fields.

Single write owner. The approval path re-enters the orchestration flow's Dataverse write action rather than writing to Dataverse independently, keeping one owner for the Dataverse write and the ledger-after-row-ID rule. It does not re-invoke the Function or Content Understanding: the fields were already extracted and corrected by the reviewer.

No isolation between spine and review. Under the Logic-Apps spine, a Power Automate outage delayed only notifications. Now that the spine is also Power Automate, a Power Automate service degradation stops invoice processing as well — plan alerting and a manual fallback (see risks).

Review queue item (SharePoint list) fields

Field	Notes
invoice_file_url	Link to the original PDF
document_type	general_invoice / property_tax / other
is_handwritten	Drives field highlighting in the review app
extracted_fields / confidence_scores	Full CU payload and per-field scores
rejection_reason	Which gate fired (A1, B2–B4, GST)
anomaly_flag	Advisory — shown to the reviewer
status	Pending / Approved / Rejected

8. Key risks and watch items

Severity	Item
RED	Duplicate invoices are not detected (requirement): a re-uploaded or manually-entered invoice creates an additional Dynamics row. Gate A1 only prevents reprocessing the same upload.
RED	CE vendor lookup data model: invoices reference accounts via lookup, not free text; a 'find or create account' step may precede the write. Confirm with the CE admin before Phase 6.
RED	Optimistic ledger writes: the terminal Written status and row ID are written only after Dataverse returns 201. An empty DynamicsRecordId marks a failed or in-progress write.
RED	Handwriting is the accuracy and fraud cliff: is_handwritten = yes routes to mandatory review, no exceptions at launch.
AMBER	Power Automate is now the spine, not just review: a Power Automate service degradation stops the whole pipeline. The isolation the Logic-Apps spine gave is gone — add alerting and a manual fallback.
AMBER	Power Automate auth surface: the HTTP action has no managed identity, so the Function call uses a custom connector (function-key connection now; Entra federated-credential managed identity for production). Dataverse and SharePoint use connection references — credential ownership and rotation are part of the security surface.
AMBER	Power Automate throttling, concurrency limits, and per-action request timeouts apply to the spine. The Function call is synchronous, so CU latency must stay under the connector timeout; if it nears the limit, move the Function to the async 202 + polling pattern.
AMBER	Concurrent trigger replays: A1's atomic insert-claim makes the same upload claim-once; a row stuck at Received is resumed only after the staleness lease (default 600s).
AMBER	Multi-page line-item tables: a known CU limitation can drop earlier-page rows; the GST check routes inconsistent arithmetic to review.
GREEN	Region: West US validated, all resources co-located. Canada Central is incompatible.